Function as a Service (FaaS) Support for the HYDRA Resource Broker

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Abstract

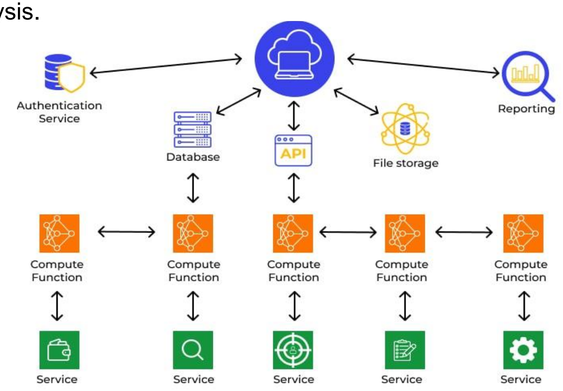
Modern scientific workflows increasingly require on-demand and event-driven task execution across both Cloud and High-Performance Computing (HPC) environments. To meet this need, we extended HYDRA-—a resource broker originally developed to manage heterogeneous workloads across HPC and Cloud platforms—-to support Function-as-a-Service (FaaS) capabilities on the cloud side. This enhancement was implemented at Computational Science Department (Brookhaven National Laboratory), where we developed two core components: a lightweight FaaS Agent that interfaces with Nuclio, an open-source platform extending Kubernetes to support data-intensive serverless workloads; and a FaaS Manager that orchestrates the Agent along with AWS Lambda provider modules responsible for deploying and invoking standalone modules, packages, or inline functions.

To quantify system performance and overhead, we conducted a series of benchmarks using diverse workloads, including CPU-intensive image processing, I/O-intensive operations with AWS S3, and high-frequency burst functions. These tests evaluated key metrics such as end-to-end invocation latency, manager processing overhead, and cold start penalties in event-driven scenarios. Scalability tests with high concurrency levels were also performed to examine system responsiveness and elasticity under load. The results demonstrate that the FaaS Manager introduces minimal overhead—averaging under 3 seconds when using the Lambda provider, and scales efficiently, validating its suitability for integrating serverless computing into complex scientific workflows.

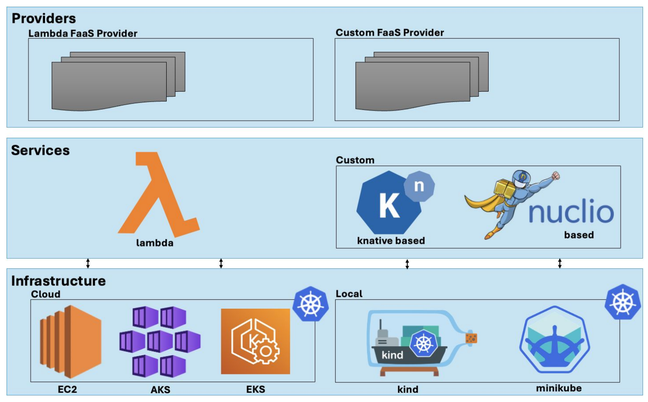
This project aligns with the Computational Science Department's mission to deliver reliable, high-performance software infrastructure for a broad range of scientific domains. Through this work, I gained experience in Python development, Continuous Integration using GitHub, serverless computing technologies, containerization with Docker and Kubernetes, and performance evaluation methodologies for computational research.

Function as a Service (FaaS)

Function as a Service (FaaS) is a cloud computing model where you run small, single-purpose pieces of code without managing any servers. This "serverless" approach means that code executes only when triggered by an event, and you only pay for the exact time it runs. For scientific workflows, FaaS is useful for automatically processing new data, running thousands of small, parallel calculations, and connecting different steps in a complex analysis.



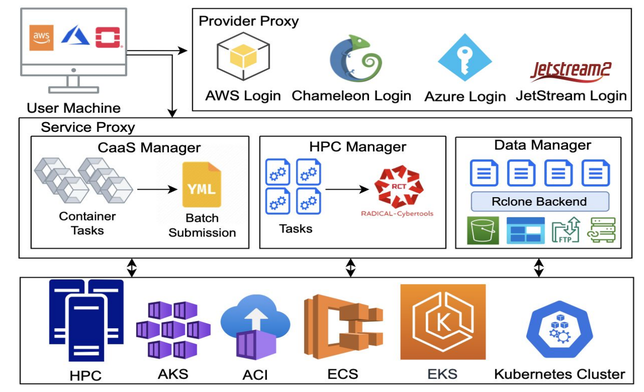
Introduction



To address the growing need for running small, on-demand tasks in scientific research, we developed a **FaaS Manager** that extends the capabilities of the **HYDRA** system. This manager provides a single, unified hub to deploy and run code as functions on different platforms, hiding the complexity of underlying cloud services. It supports both the widely used **AWS Lambda** and the high-performance, open source **Nuclio** platform, with the **Nuclio** module being specially optimized for data-intensive work.

HYDRA

Hydra is a resource brokering system that helps scientists run complex computational workloads across diverse platforms like High-Performance Computing (HPC) clusters and various Cloud providers. It acts as a manager, automatically acquiring resources and executing tasks on behalf of the user asynchronously. This simplifies using different computing environments together, letting researchers focus on their science instead of system management



HYDRA FaaS Workflow Architecture

Orchestration Workflow: This diagram shows a high-throughput FaaS (Function-as-a-Service) architecture for deploying serverless functions across multiple platforms.

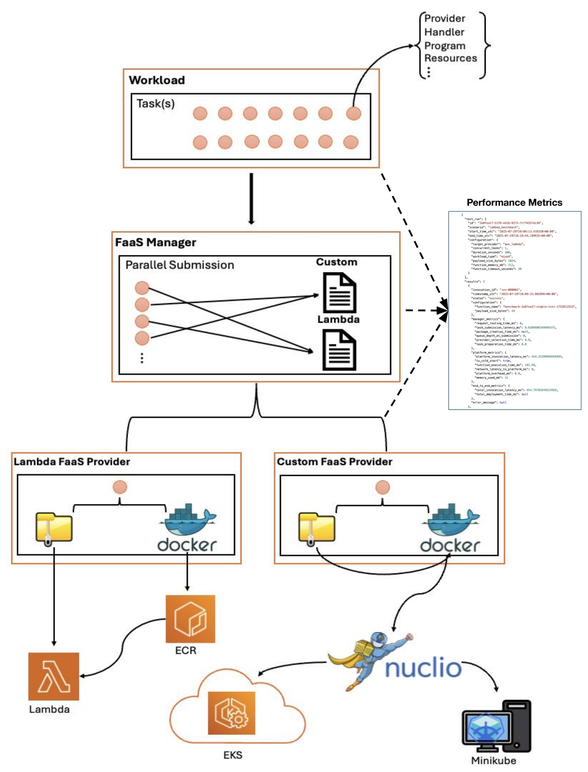
Central Manager: A central Manager receives tasks and uses a "Parallel Submission" mechanism to concurrently route them to the appropriate provider based on the task's configuration.

Lambda Provider: This provider handles deployments to AWS Lambda. It can either package the function as a ZIP file or build a container image that it pushes to Amazon ECR before deployment.

Custom (Nuclio) Provider: This provider deploys functions to the Nuclio serverless platform. It supports deploying to various Kubernetes environments, including cloud-based clusters like Amazon EKS or local setups like Minikube.

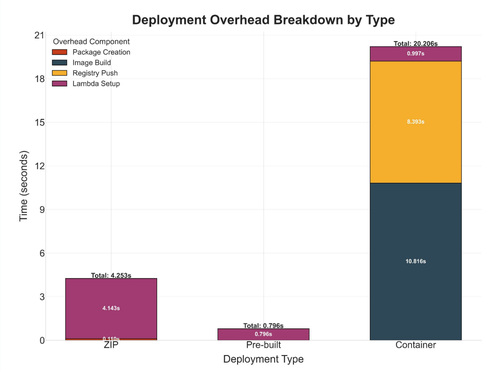
Keywords

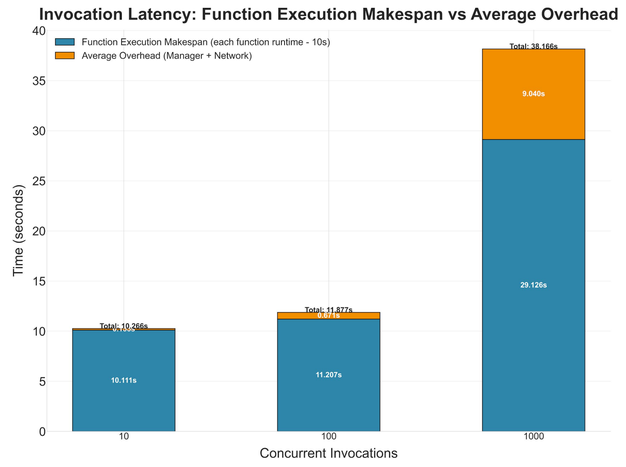
- AWS EC2 (Elastic Compute Cloud): A remote server in the cloud.
- AWS ECR (Elastic Container Registry): A container image store.
- AWS EKS (Elastic Kubernetes Service): A a container orchestrator.
- AWS Lambda: Serverless function platform.
- Docker: Containerization technology.
- Minikube: A Kubernetes environment for testing.

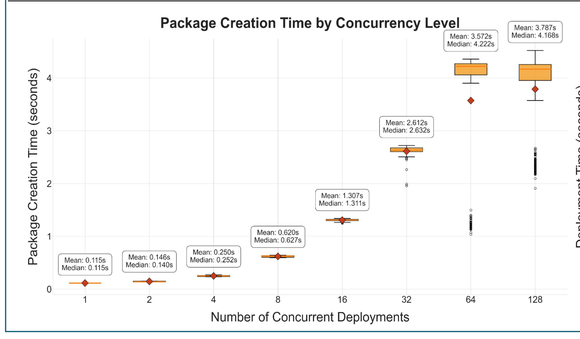


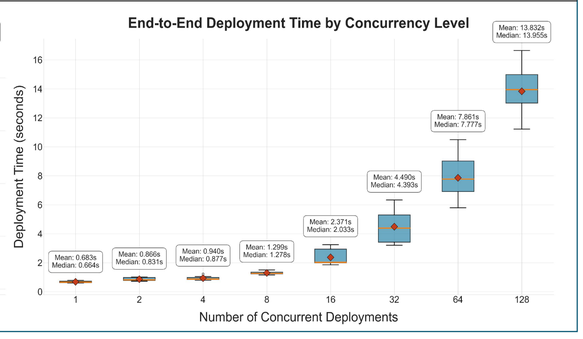
Results

Resource utilization (avg):
CPU (1 core) - 47.9%, **Memory** - 272 MB
Resource provisioning times (avg, sec):
IAM - 10.3±0.1, **ECR** - 0.3









Conclusion

The overarching goal of this project was to extend the HYDRA Resource Broker with Function-as-a-Service (FaaS) capabilities, enabling it to manage the on-demand, event-driven tasks required by modern scientific workflows. Our first implementation focused on creating a provider for AWS Lambda, a widely used commercial service. Although this provided a straightforward path for deploying simple functions, it was ultimately an incomplete solution, as it lacked a control to gauge its performance.

At last, a comprehensive solution was achieved by developing a second, high performance provider for the open-source Nuclio platform and a FaaS Manager to orchestrate both. With the addition of performance optimizations like batch processing to the Nuclio module, the system consisting of both an accessible commercial provider and a powerful self hosted one now provides a robust and flexible FaaS capability for HYDRA. Performance tests confirm this final design operates with negligible overhead, making it a truly practical tool for research.

As this work continues in the future, the FaaS Manager and its new capabilities will be integrated into the Framework for Assessing Changes to Sea Level (FACTS). A functional early prototype of the FaaS Manager has already demonstrated successful execution of diverse workloads in a serverless environment. This prototype will serve as the foundation for integration into FACTS, enabling climate scientists to run components of their sea level projections as scalable, on-demand functions, and to characterize uncertainty in climate predictions and advance our understanding of sea level change.

Acknowledgments

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